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HTML

Markup languages

- Markup language by specific items (markups) give a meaning to enclosed elements
- Distinguishes the content from its meaning
- A simple form, however redundant
- A few examples:
 - XML
 - HTML
 - Latex
 - MathML, SVG

Markup languages

XML (

```
<xsd:element name="P_39" type="etd:TKwota2Nieujemna">
  <xsd:annotation>
    <xsd:documentation>3. Ulga na dzieci</xsd:documentation>
  </xsd:annotation>
</xsd:element>
```

HTML (zut.edu.pl)

```
<li class="condensed">
  <a href="/EN/university.html" style="pointer-events: auto;">
    <span class="lang-top-ns">EN</span>
    <span class="sr-only">English version</span>
  </a>
</li>
```

LaTeX (nyu.edu)

```
\begin{equation} % Creates an equation environment and is compiled as math
  \gamma^2+\theta^2=\omega^2
\end{equation}
```

Tag

- A tag is an HTML entity used to create an element, with unique name (selected according to the HTML standard) enclosed by `<` and `>` characters
- `<tag_name>` and `</tag_name>`
- The names of tags make their understanding easier (short & easy to remember)

Tag

- There are 3 scenarios relating to usage of tags:
 - Usage of opening/starting & closing/ending tags
`<tag_name>... .</tag_name>`
 - Opening/starting tag (discussed later) not closed
`<tag_name>`
 - Void element with tag closed by the same item
`<tag_name/>`

Attribute

- An HTML attribute is additional value that configures element
- Provided as a name withing a tag
`<tag attribute_name="attribute_value">`
or
`<tag attribute_name='attribute_value'>`
- Be consistent with usage of single and double quotes!

Content

- A content of HTML tag is information enclosed within opening and closing tags

```
<tag>THIS_IS_A_CONTENT</tag>
```

- It can be a text to display
- It can also be a nested other tag
- Nesting of elements can be continued

A map of combinations

	No-content	With content
No-attribute(s)	<code>
</code>	<code><i>Text in italics</i></code>
With attribute(s)	<code></code>	<code>The other page</code>

- The above map covers all elementary combinations of tags-attributes-content
- An assignment is not constant i.e. tags do not always fall into the same cell

Additional notes - essential for devs

- Use lower case letters (HTML allows to use both) – cleaner code, standard in the community
- Close tags – makes processing easier; no significant cost – might be very beneficial in the future
- Use comments – essential during development, can be removed at the production

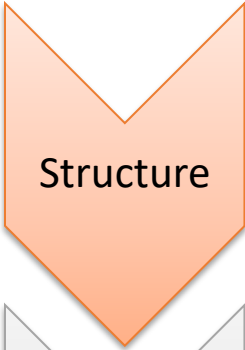
Additional notes – continuation

- Quote attributes even if HTML allows attribute values to be unquoted
- Do not use spaces around equal signs i.e. `attribute = "value"`
- Split long lines of code – the browser manages to render it, but its an issue for developers
- Use formatting (blank lines and indentation) – manual, provided by IDE or formatting tools

Additional notes – continuation

- Use lower case in references – case sensitivity depends on OS and configuration of Web server
- Select one .html (common choice) or .htm and do not add extra overriding rules at the server
- Try to be as standardised as possible

HTML – Groups of tags #1 Classification



Structure

- Define the structure of page
- Basic type in HTML



Presentation

- Define presentation of elements
- Auxiliary group, mainly replaced by CSS



Semantics

- Give meaning to the elements
- A rare usage

HTML – #2 Classification (MDN) part 1

Main root

- `<html>`

Metadata

- `<base>`, `<head>`, `<link>`, `<meta>`, `<style>`, `<title>`

Sectioning root

- `<body>`

Content sectioning

- `<address>`, `<article>`, `<aside>`, `<footer>`, `<header>`,
`<h1>-<h6>`, `<main>`, `<nav>`, `<section>`

Text

- `<blockquote>` `<dd>`, `<div>`, `<dl>`, `<dt>`, `<figcaption>`,
`<figure>`, `<hr>`, `<li>`, `<menu>`, `<ol>`, `<p>`, `<pre>`, `<ul>`

HTML

HTML – #2 Classification (MDN) part 2

Inline semantics

- <a>, <abbr>, , <bdi>, <bdo>,
, <cite>, <code>, <data>, <dfn>, , <i>, <kbd>, <mark>, <q>, <rp>, <rt>, <ruby>, <s>, <samp>, <small>, , , <sub>, <sup>, <time>, <u>, <var>, <wbr>

Image & multimedia

- <area>, <audio>, , <map>, <track>, <video>

Embedded content

- <embed>, <iframe>, <object>, <picture>, <portal>, <source>

SVG and MathML

- <svg>, <math>

Scripting

- <canvas>, <noscript>, <script>

Demarcating

- , <ins>

HTML – #2 Classification (MDN) part 3

Table

- `<caption>`, `<col>`, `<colgroup>`, `<table>`, `<tbody>`, `<td>`, `<tfoot>`, `<th>`, `<thead>`, `<tr>`

Forms

- `<button>`, `<datalist>`, `<fieldset>`, `<form>`, `<input>`, `<label>`, `<legend>`, `<meter>`, `<optgroup>`, `<option>`, `<output>`, `<progress>`, `<select>`, `<textarea>`

Interactive

- `<details>`, `<dialog>`, `<summary>`

Web components

- `<slot>`, `<template>`

Obsolete and deprecated

- For a list look at MDN reference; Avoid from now on and replace in existing code

References

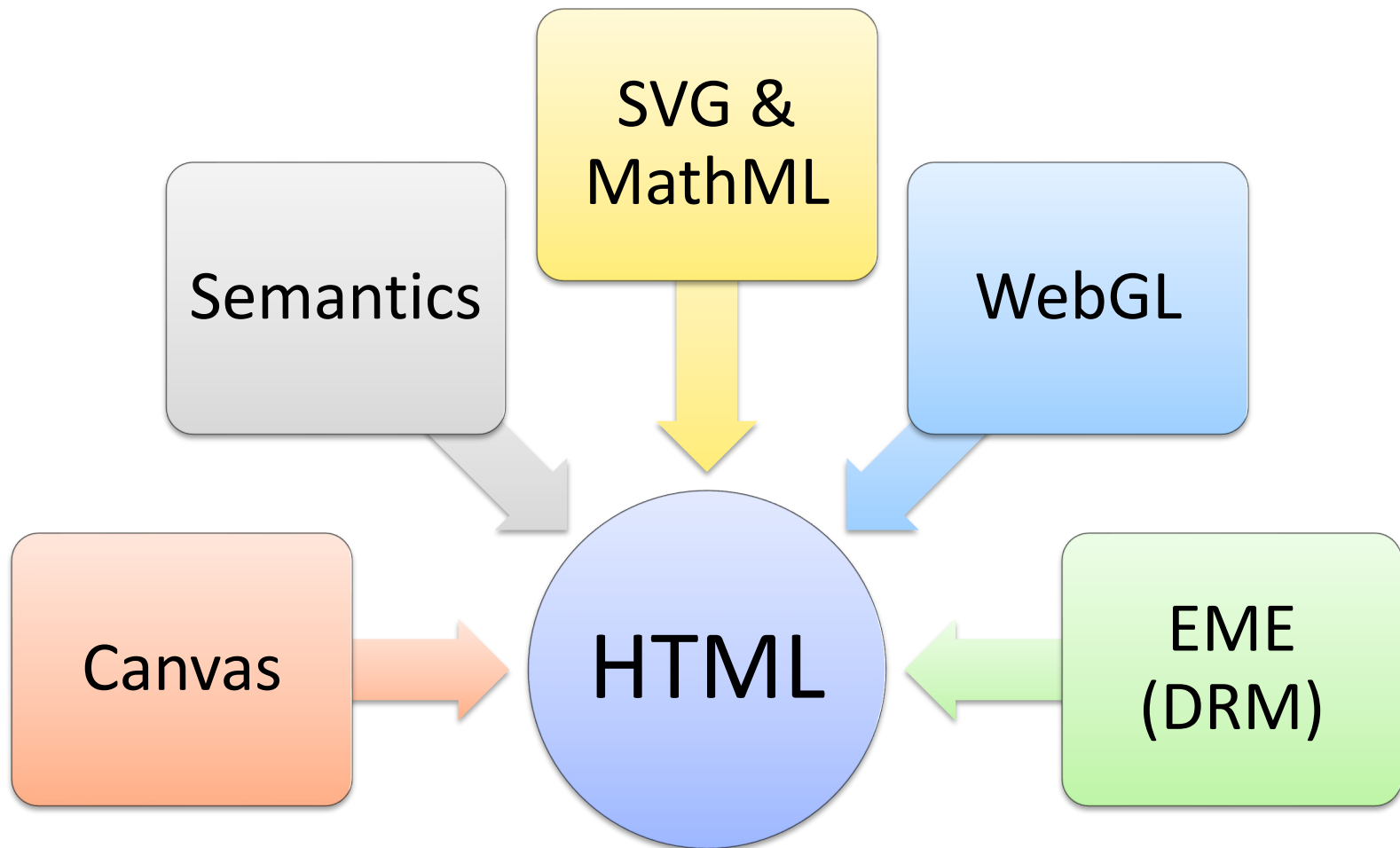
- For tags use:
<https://developer.mozilla.org/en-US/docs/Web/HTML/Element>
- For attributes use:
<https://developer.mozilla.org/en-US/docs/Web/HTML/Attributes>
- or any other documentation (should follow the web standards)

Special entities in HTML

- Some special characters have encodings, e.g.:
 - Space ` `; ` `
 - `<` `<`; `<`
 - `>` `>`; `>`
 - `"` `"`; `"`
 - `'` `'`; `'`
 - `&` `&`; `&`
 - © `©`; `©`
 - ® `®`; `®`
 - ™ `™`; `™`
- 
- A diagram consisting of a large right-facing curly bracket on the left side of the list, spanning from the '<' entity to the '&' entity. To the right of the bracket is a green rectangular box with a thin border containing the text 'Special characters'.

See <https://dev.w3.org/html5/html-author/charref>

HTML5 – Additional elements



HTML5 - Tags

- The current documentation of HTML5 is available at:
<https://html.spec.whatwg.org/multipage/>
- Browser compatibility data for HTML, CSS, JS, SVG is available at:
<https://github.com/mdn/browser-compat-data>

xHTML

- xHTML is an XML application, resembling HTML – HTML meaning, with XML rules
- Core changes to HTML:
 - It is necessary to close tags
 - Closing must be in order
 - Empty elements must be closed
 - Lower letters must be used in tags and attributes

XHTML

- XHTML is an XML application with all of its benefits e.g.:
 - Ability to use XML tools and technologies (XSL Transformation, XML Schema etc.)
 - Integration with other XML applications
- and issues:
 - More complex (more to learn)
 - Rigorous rules (more to remember about)
- In 2021 7% of websites uses it (W3Techs.com)

SOFTWARE & TEMPLATES

Tools for HTML and CSS (next lecture)

Code editors	• Notepad++, Geany, Sublime Text, Visual Code Studio, Atom
Front-end frameworks	• Bootstrap, HTML5Boilplate, ChromeDevTools, Semantic UI, Materialize
Web-app frameworks	• Django, Angular, React, Vue, Node.js, Spring
Package managers	• npm, DPKG, Yarn, Pip, yum, maven, ant
Version control	• git command line, GitHub Desktop, Sourcetree
Testing tools	• Advanced REST Client, REST Assured, Postman
Prototyping tools	• Figma, Sketch, ProtoPie, Adobe XD, MockFlow, Visual Paradigm, Origami Studio
Collaboration tools	• JIRA, Asana, Slack, Zoom
Miscellaneous	• TypeScript, CSS Pre-processors, Image Tools

Templates #1

- Don't (if not desired) start development from a empty file – use a template/boilerplate code
- Lower cost of development
- Faster development
- Better quality (initially – do not spoil it):
 - Conformance to standards
 - Proper Structuring
 - Responsiveness

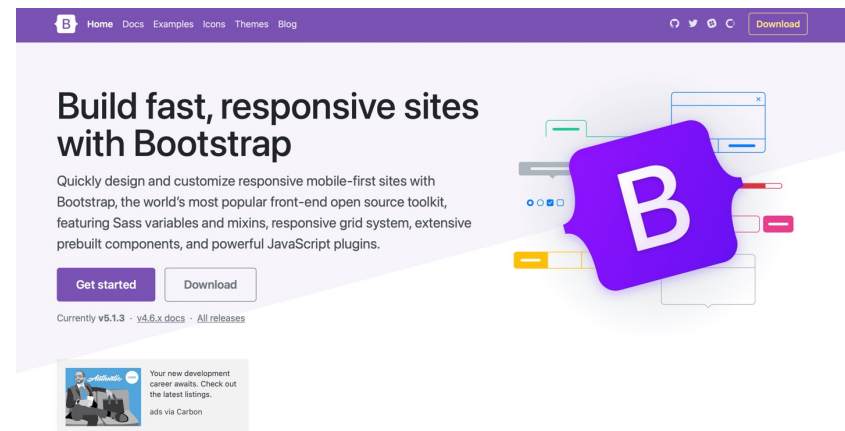
Templates – HTML5Boilerplate

- The most popular HTML5 template
- Focus on best practice site-configuration
- Cross-browser compatibility
- More generic template
- Open to the other components



Templates - Bootstrap

- Another number #1
- Starting-up project with base HTML, CSS, JS, forms, buttons, tables, navigation, grids, icons etc.



Summary

- HTML syntax is easy
- HTML semantics is easy to understand
- HTML5 covers many additional technologies
- It is possible to start with a scratch, however, isn't necessary (at least it shouldn't be)
- Reuse elements – especially the ones developed by 100s of others (wisdom of crowds)

TILL NEXT LECTURE